

ELEC 473 – Cryptography and Network Security

Quiz 2

Friday, Nov. 04, 2022, 8:45-9:20 AM

INSTRUCTIONS—Read Carefully

- Write your name and ID legibly at the top of the papers.
- Calculators are allowed to use.
- Show all your work. Partial credit will be given for substantial progress towards the solution. No credit will be given for answers with no substantial process.

Part I Multiple Choices - please mark the best answer (subtotal: 10 marks).

1. Suppose a prime $p \equiv 3 \pmod{4}$ and $a \in \mathbb{Z}_p$. Which of the following is equivalent to a square root of $a \pmod{p}$?

- A. $a^{p-1} \pmod{p}$
- B. $a^{(p+1)/4} \pmod{p}$
- C. $a^{(p+1)/2} \pmod{p}$
- D. $a^{(p-1)/2} \pmod{p}$

2. Let $C(K, M)$ denote a message authentication code function, produced for the message M and a shared key K . Let $E(K, M)$ denote encryption of a message M with a key K , and let $||$ denote the concatenation. If Alice sends to Bob the following information: $E(K_2, M || C(K_1, M))$ where K_1 and K_2 are shared secret keys, it is

- A. just message authentication
- B. message authentication and confidentiality where authentication is tied to the plaintext
- C. message authentication and confidentiality where authentication is tied to the ciphertext
- D. message authentication and confidentiality where authentication is tied both to the plaintext and to the ciphertext

3. A receiver must be able to check a message to prevent the adversary from modifying it during transmission, it is known as

- A. Message integrity
- B. Message non-repudiation
- C. Message confidentiality
- D. Message sending

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4. The elliptic curve group is defined by $y^2 = x^3 - 17x + 16$ over real numbers, what is $P + Q$, if $P = (0, -4)$ and $Q = (1, 0)$?

A. (15, -56)

B. (-23, -43)

C. (69, 26)

D. (12, -86)

$$\lambda = \frac{y_2 - y_1}{x_2 - x_1}$$
$$x_3 = \lambda^2 - x_1 - x_2$$
$$y_3 = (x_1 - x_3)\lambda - y_1$$

5. Which of the following statements is incorrect?

A. The large integer n factorization problem is equivalent to the problem of computing $\varphi(n)$ from n . (There is a typo, if you choose A, you will not lose mark)

B. If an adversary can factor the large integer n , it can address the RSA hard problem.

C. The security of RSA is based on the hard problem of large number factorization.

D. If an adversary can address the RSA hard problem, it can factor the large integer n .

6. Which one of the following algorithms does not belong to asymmetric-key cryptography?

A. RSA algorithm

B. Diffie-Hellman Key Agreement algorithm

C. Message Authentication Code algorithm

D. None of the mentioned

7. Which of the following is an advantage of public-key cryptography over symmetric-key cryptography?

A. Public-key cryptography provides more security services

B. Public-key cryptography does not rely on conjectured hardness of certain computational problems

C. Public-key cryptography has higher throughput

D. Public-key cryptography has shorter key size

8. Consider the following system of two equations (congruences):

$$x \equiv 12 \pmod{29}$$

$$x \equiv 7 \pmod{15}$$

According to Chinese Remainder Theorem (CRT), which of the following is true about x ?

A. x has no solution

B. x has exactly one solution

C. x has more than one solution

D. x has four solutions because there are two equations

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9. What is the advantage of elliptic curve cryptosystems compared to the RSA and Diffie-Hellman based cryptosystems?

A. Short key size

B. Fast encryption and decryption

C. Secure against Quantum attacks

D. None of the mentioned

10. What is the order of Elliptic Curve group $(E,+)$ over Z_p ?

A. $p-1$.

B. The number of points on $E(Z_p)$.

C. The number of points on $E(Z_p) + 1$.

D. $p+1$.

Part II (subtotal: 20 marks).

Q1.(10 marks) Solving the following system of congruences:

$$3x \equiv 1 \pmod{7}$$

$$9x \equiv 5 \pmod{11}$$

$$7x \equiv 5 \pmod{13}$$

$$\text{GCD}(7,11)=1 \quad \text{GCD}(7,13)=1 \quad \text{GCD}(11,13)=1$$

$$\begin{aligned} 3^{-1} \pmod{7} &= 5 & 9^{-1} \pmod{11} &= 5 & 7^{-1} \pmod{13} &= 2 \\ \begin{cases} 3x \equiv 1 \pmod{7} \\ 9x \equiv 5 \pmod{11} \\ 7x \equiv 5 \pmod{13} \end{cases} &\Rightarrow \begin{cases} 3 \cdot 3x \equiv 3 \cdot 1 \pmod{7} \\ 9 \cdot 9x \equiv 9 \cdot 5 \pmod{11} \\ 7 \cdot 7x \equiv 7 \cdot 5 \pmod{13} \end{cases} &\Rightarrow \begin{cases} x \equiv 5 \pmod{7} \\ x \equiv 3 \pmod{11} \\ x \equiv 10 \pmod{13} \end{cases} \end{aligned}$$

$$M = 7 \times 11 \times 13 = 1001$$

$$M_1 = 143, \quad M_2 = 77, \quad M_3 = 77$$

$$y_1 = 143^{-1} \pmod{7} = 5, \quad y_2 = 77^{-1} \pmod{11} = 4, \quad y_3 = 77^{-1} \pmod{13} = 12$$

$$\begin{aligned} x &= 5 \times 143 \times 5 + 4 \times 77 \times 3 + 12 \times 77 \times 10 \pmod{1001} \\ &= 894 \pmod{1001} \end{aligned}$$

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Q2. (10 marks) Let E denote the elliptic curve $y^2 \equiv x^3 + x + 6$, defined over Z_{11} . It can be shown that $\#E=13$. Considering the Elliptic Curve ElGamal cryptosystem that is implemented in E . $P=(2, 7)$ is the generator of $E(Z_{11})$ and $a=5$ is the private key of Alice.

- Compute the public key of Alice.
- Suppose Bob needs to send a message $m=(3, 5)$ to Alice. Bob chooses a random value $k=7$ for encryption. Compute the corresponding ciphertext of m .
- Show the decryption of the ciphertext in b from Bob.
- If the message $m=(10, 2)$ and the random value $k=3$, what is the ciphertext?

(Hint: $P=(2,7)$, $2P=(5,2)$, $3P=(8,3)$, $4P=(10,2)$, $5P=(3,6)$, $6P=(7,9)$, $7P=(7,2)$, $8P=(3,5)$, $9P=(10,9)$, $10P=(8,8)$, $11P=(5,9)$, $12P=(2,4)$)

- $PK=aP=5(2, 7)=(3, 6)$
- $U=kP=7(2, 7)=(7, 2)$
 $V=m+kPK=(3, 5)+7(3, 6)=(3, 5)+(10, 9)=(10, 2)$
- $m=V-aU=(10, 2)-5(7, 2)=(10, 2)-(10, 9)=2(10, 2)=(3, 5)$
- $U=kP=3(2, 7)=(8, 3)$
 $V=m+kPK=(10, 2)+3(3, 6)=(10, 2)+(5, 2)=(7, 9)$